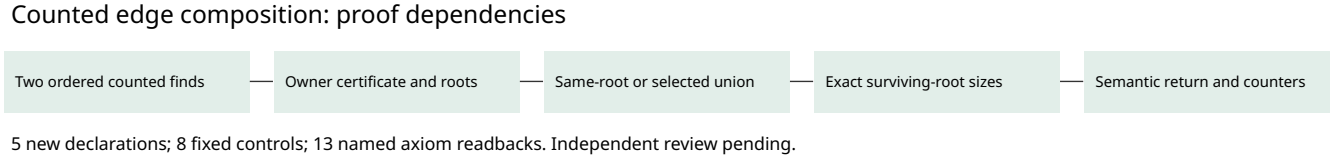
Counted signed edges preserve component invariants

Five new declarations and eight fixed controls compiled. Thirteen separate named axiom readbacks passed; the only listed axioms are propext, Classical.choice and Quot.sound. Three malformed inventories are rejected, and Lean rejects the deliberately false claim that a same-root edge changes its owner. Independent fidelity review is pending publication of this producer packet.



The owner proof follows two ordered word finds: first s to t, then t to w. Decoded find lemmas derive the retained owner certificate and both surviving roots. Same-root dispatch keeps the graph and sizes. Distinct-root dispatch applies the existing attachment theorem in the native size-selected orientation, deriving merged owner fibers and exact root cardinalities. Output owner correctness and output cardinalities are conclusions, never premises.

The composition theorem connects BranchCount.run to the existing EdgeDispatch result and joins canonical potentials and anchors, signed semantic validity, forest existence, safe return, component owners and root sizes. It also proves at least two find visits and the bound rank(u)+rank(v)+2, plus cycles<=1, negative<=cycles and zero<=1. A separate theorem for the same branch arguments gives all four exact integer counter updates: operations increases by ca+cb, and the other counters by their branch deltas.

Assumptions remain explicit: prime modulus with 2 < p <= 2^63, canonical word inputs and signed labels, input forest and true incoming relation, input owner/cardinality certificates, n < USize.size, and separate UInt64 no-overflow bounds. The branch-counter bounds use conservative initial + 1 < 2^64 conditions. The rank is a supplied certificate, not measured work. The reverseSizes two-node control tests orientation only and deliberately does not instantiate SizeValid; the universal reverse branch theorem derives cardinality under its stated valid-input premises. The owner-contract fixed instance retains explicit owner/size/bound hypotheses.

The current SignedGraph source block matches native-node, edge-dispatch and branch-count source revisions byte for byte. The older goal-coverage audit predates several of those components; insertion already has its six-invariant contract and was not repeated. This new edge join closes the owner/cardinality omission in the existing dispatcher contract. Concrete native array mutation/refinement and complete collector iteration remain separate constructions; allocation, interrupted mutation and compiler behavior are outside this successful value model.

Three development failures are retained: missing prime instance in owner proof; missing prime instance/import in the aggregate; and decidable fixed-control goals containing free variables. Final proofs passed without changing their intended hypotheses. Each Lean invocation used 120 seconds, 4096 MiB and one thread. Final readback binds 30,012 recursively inventoried source/cache/sidecar inputs, unchanged before and after; earlier attempts retain their narrower inventories. Imported compiled caches were reused, without independent binary/kernel rechecking. No cohort, timing measurement, cloud work or cost-ratio update occurred.